

JEE MAIN REVISION MASTER: ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 178 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Nucleophilic Addition & Carbonyl Condensation	120 PYQs	Carbonyl Reactions
Very High	Oxidation, Reduction & Identification Tests	59 PYQs	Tests & Reductions

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Nucleophilic Addition & Carbonyl Condensation	$\text{Aldol: Requires } \alpha\text{-H} \rightarrow \alpha, \beta\text{-unsaturated carbonyl}$, $\text{Cannizzaro: No } \alpha\text{-H} \rightarrow \text{Alcohol} + \text{Carboxylate salt}$
Oxidation, Reduction & Identification Tests	$\text{Iodoform test is positive for } \text{CH}_3\text{CO-} \text{ or } \text{CH}_3\text{CH(OH)- groups} \rightarrow \text{CHI}_3 \text{ (yellow ppt)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Nucleophilic Addition & Carbonyl Condensation (Very High Yield — 120 PYQs)

Core Principle: Reversible nucleophilic additions, Aldol condensation mechanism, and Cannizzaro disproportionation of non-enolizable aldehydes.

■ **Caution / JEE Pitfall:** Benzaldehyde and Formaldehyde lack α -hydrogen atoms, so they undergo CANNIZZARO reaction with concentrated NaOH, NOT Aldol condensation!

■ **Speed Trick:** Reactivity toward nucleophilic addition: $\text{HCHO} > \text{CH}_3\text{CHO} > \text{CH}_3\text{COCH}_3$ (decreases with steric crowding and +I electron donation).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Statement I : Compound (X), shown below, dissolves in NaHCO_3 solution and has two chiral carbon atoms Statement II : Compound (Y), shown below..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "'x' is the product which is obtained by the hydrolysis of prop-1-yne in the presence of mercuric sulphate under dilute acidic medium at ..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Given below are two statements: Statement I : The condensation reaction between $\text{CH}_3\text{CH}_2\text{CHO}$ and under optimum pH will produce Statement II..." [Ans: N/A]

3.2 Oxidation, Reduction & Identification Tests (Very High Yield — 59 PYQs)

Core Principle: Silver mirror Tollens' test, Fehling's cuprous reduction, yellow iodoform precipitation, Clemmensen (Zn-Hg/HCl), and Wolff-Kishner ($\text{NH}_2\text{NH}_2/\text{KOH}$) carbonyl deoxygenation.

■ **Caution / JEE Pitfall:** Benzaldehyde reduces Tollens' reagent to give silver mirror, but does NOT reduce Fehling's solution!

■ **Speed Trick:** Clemmensen reduction is suitable for acid-stable compounds. For acid-sensitive carbonyls, use Wolff-Kishner reduction (basic medium).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Consider the following reaction. The major product(P)formed is :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Complete the following reaction sequence and give the name of major product 'P'..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "A molecule (X) with following structure under mild acidic condition is hydrolysed to produce (Y) and (Z). Identify the correct statements ab..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 22nd January Morning Shift (Q#3)

Match the LIST-I with LIST-II List-I Reagents List-II Name of Reaction involving carbonyl compounds A.

$\text{NH}_2\text{NH}_2\text{KOH}$ I. Tollen's Test B. $\text{Ag}(\text{NH}_3)_2\text{OH}$ II. Clemmensen Reduction C. Aq. CuSO_4 , Sodium Potassium tartarate, KOH III. Wolff - Kishner Reduction D. Zn-Hg,HCl IV. Fehling's Test Choose the correct answer from the options given below :

- (A) -IV, B-III, C-II, D-I
- (B) -III, B-I, C-IV, D-II [Correct]
- (C) -II, B-I, C-IV, D-III
- (D) -III, B-IV, C-I, D-II

Correct Answer: B

Solution Analysis: A reagent is $\text{NH}_2\text{NH}_2\text{KOH}$ (hydrazine in strong base). This is used to reduce aldehydes/ketones to hydrocarbons by Wolff–Kishner reduction . So, A $\rightarrow$ III . B reagent is $\text{Ag}(\text{NH}_3)_2\text{OH}$ (ammoniacal silver nitrate). This gives silver mirror with aldehydes: Tollen's test . So, B $\rightarrow$ I . C reagent is aqueous CuSO_4 + sodium potassium tartrate (Rochelle salt) + KOH ; together they form Fehling's solution , used in Fehling's test for aldehydes (red Cu_2O ppt). So, C $\rightarrow$ IV . D reagent is Zn – Hg,HCl which reduces carbonyl group to $-\text{CH}_2-$ in ac

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Evening Shift (Q#4)

Given below are two statements : Statement I : Cross aldol condensation between two different aldehydes will always produce four different products. Statement II : When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH , it forms a condensation product with acetophenone only. In the light of the above statements, choose the correct answer from the options given below

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are false **[Correct]**
- (C) Both Statement I and Statement II are true
- (D) Statement I is false but Statement II is true

Correct Answer: B

Solution Analysis: Statement I: False. A crossed aldol between two different aldehydes gives four products only in the uncontrolled case when both aldehydes have $\alpha\text{-H}$ (2 self-aldols + 2 crossed). It is not "always" : e.g., benzaldehyde (no $\alpha\text{-H}$) cannot self-aldol, so fewer products result; and under controlled conditions you can also avoid mixtures. Statement II: False. Semicarbazide forms semicarbazones with both aldehydes and ketones , and aldehydes (benzaldehyde) generally react faster than ketones (acetophenone) under the usual "optimum pH" conditions. So it will not give a condensation product with acetopheno